

What Factors Matter for Financial Development in South Asian Emerging Developing Countries: A System Generalized Method of Moment (SGMM) Analysis

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Abstract

This paper aims to assess the factors which determine financial development in South Asian emerging countries over the period of 1976 to 2013. We apply System Generalized Method of Moment (SGMM) as well as pooled OLS and Fixed Effects Model (FEM). Results indicate that maintaining a lower level of inflation is most important determinants of all proxies of financial development in South Asian countries. Results also exhibit that remittance inflows are directly and positively associated with financial development. We also find that economic growth and trade openness impact positively financial development through generating demand for more and new financial services which result in establishing new financial intermediaries capable of meeting new demand for services. Policies leading to maintaining lower inflation rate, encouraging remittance inflows, generating economic growth would have boasting impact on financial development which in turn could have accelerating effect on economic growth in South Asian emerging countries; and thus keeping trade liberalized and easing the flow of remittances should be put in the policy options.

Keywords: Financial Development, Pooled OLS, Fixed Effects Model (FEM) and System Generalized Method of Moment (SGMM)

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1. Introduction

A well-developed financial system is instrumental in attaining sustainable and balanced growth. Such a system increases the availability of funding by mobilizing idle savings, facilitating transactions and attracting foreign investments. Such a system can achieve improved allocation of financial resources and enhanced risk management, transparency and corporate governance practices. Moreover, developed financial systems can ease the availability of some credit to more opaque businesses, such as first-time or low-income (and low collateral) borrowers or small- and medium-sized enterprises (SMEs). In short, financial development can serve to improve not only growth prospects but also distribution of economic opportunities.

The main issues of financial development raise some important questions, i.e., why some countries have so much bigger capital markets as compared to others, e.g., why countries like the United Kingdom and United States have huge financial markets, whereas countries like France and Germany have comparatively smaller financial markets. Why Japan and Germany have enormous banking systems as compared to the other wealthy economies. There are some significant reasons for the different levels of development in different countries.

A strong financial system offers risk diversification and effective capital allocation. The greater the financial development, the higher would be the mobilization of savings and its allocation to high return projects. Financial development could be explained by the access to credit and financial services (Levine 1997, 2005), can accelerate growth (Schumpeter, 1912), simplifies the “channels” of economic growth (Robinson, 1952), promotes the economic growth (Miller, 1998 and Lucas, 1988), and has had a positive relationship with economics growth (Goldsmith, 1969; McKinnon, 1973; La Porta et al., 1998; Levine et al., 2000; and Beck et al., 2000)

South Asian emerging countries (i.e., Bangladesh, India, Pakistan and Sri Lanka) started the financial liberalization process in the early 1990’s (Bekaert, Harvay and Lundblad, 2001). This region is important since aggregate net flows to this region have increased since the mid 1980’s (Hussain and Jun, 1992). We choose these countries on the basis of similarity of proxies of financial development and availability of balanced panel data. According to Global Development Finance (2006), private capital flows to South Asia have more than doubled since

2000. For example, capital flows reached \$23.6 billion in 2005, compared to \$9.7 billion in 2000. In South Asia, Foreign Direct Investment (FDI) increased to \$8.4 billion, representing an increase of \$1.2 billion since last year. The report mentions that India received a major share of capital flows in South Asia. Liberalization efforts in the 1990's as well as the subsequent surge in capital flows to South Asian emerging countries make them especially interesting to study.

The higher the degree of financial development, the wider is the availability of financial services. A developed and structured financial system offers higher returns with less risk and hence can boost an economy. The main question now is how to measure financial development. Measures of financial development like real interest rates and ratios and broad money to GDP can lead to misleading results (Pill and Pradhan, 1995). These indicators overlook the financial openness of a country and ignore figures of public borrowings which are made from domestic financial systems.

Antzoulatos et al (2008) includes banks, financial institutions, stock market and bond market development to construct financial development indices. The proxies for banking sector development include deposit money bank assets to GDP, bank overhead costs, bank's concentration, bank's net interest margin, and private credit issued by domestic money bank and other financial institutions to GDP. The proxies for the development of financial institutions include life insurance premium and non-life insurance premium. The development of stock market is captured by the stock market capitalization to GDP, stock market total value traded to GDP, and turnover ratio of stock market. The proxies for the bond market development include private bond market capitalization to GDP and public bond market capitalization to GDP. The ratio of bank deposit liabilities to GDP is used to measure the financial development (Demetriades et al, 1996).

This study chooses domestic credit provided by financial sector, domestic credit to private sector and broad money as a percentage of GDP as indicators of financial development as these are available for all South Asian emerging countries. This paper aims to assess determinants of these indicators of financial development for applying SMMM methods along with pooled OLS method and FEM method. Bond, Hoeffler and Temple (2001) argue that the pooled OLS and the FEM estimators should be considered respectively as the upper and lower bound for the system GMM coefficients.

The remainder of the paper is organized as follows. Section 2 provides a brief literature review; Section 3 describes determinants of financial development; Section 4 gives specification of the empirical model; Section 5 details data and methodology; Section 6 deals with econometric methodology; Section 7 explains estimated results and Section 8 gives a conclusion.

2. Literature Review

Drrucci et al (2009) measure domestic financial development in 26 emerging economics and report that different levels of domestic financial development tend to be associated with the building up of external imbalances across countries. Applying system Generalized Method of Moments (SGMM) method, Huang (2006) reports positive effect of private investment on financial development for 43 countries. In analyzing relationships among legal and institutional development, financial development and capital account liberalization using a panel data for a sample of 108 countries over 1980 to 2000, Chinn et al (2006) report that a higher level of financial openness contributes to the growth of equity markets.

Huang (2005) studies determinants of financial development applying jointly Bayesian Model Averaging and General-to-Specific approaches and suggests that institutional quality, geographic characteristics and macroeconomic policies, level of cultural and earning characteristics are associated with financial growth. Baltagi et al. (2007) show that trade openness and financial openness together with economic institutions determines the financial development. Quartey (2005) documents that financial sector development does not Granger-cause savings mobilization in Ghana and that there is a long-run cointegration relationship between financial sector development and poverty reduction.

Therefore, we see that there are several studies on determinants of financial development applying different methodologies on different countries across the world. But no studies are found on South Asian emerging countries that employ static panel data method to determine the financial development. This study fills the gap and indentifies the determinants of financial development in South Asian emerging countries.

3. Determinants of Financial Development

The economics literature has identified a number of determinants that contribute to development of financial sector. Among these determinants, the institutional and regulatory preconditions are

perhaps the most studied factors underlying a well-functioning financial system. Studies report that financial development is stronger when institutions that protect and match the needs of investors are present. Empirical evidence shows that reinforcing the rights of creditors and contract enforcement tends to deepen financial markets (Law and Azman-Saini, 2008). Availability of information on borrowers improves the supply of credit and enhances efficiency of financial institutions, especially in less developed financial sectors (Djankov et al., 2007). Lastly, institutional quality, in particular the extent to which institutional checks and balances exist, are also found to be crucial in determining the success of financial reforms and mitigating the likelihood of crises (Tressel and Detragiache, 2008). Pagano and Volpin (2001) report that countries with closed and static political regimes tend to resist the availability of external financing, since the ensuing competition would threaten the entrenched powers of the political elite. Rajan and Zingales (2003) illustrate that these pressures may weaken as economies open up to international trade and finance, resulting in the subsequent development of financial systems. Using a firm-level dataset of a large number of Pakistani firms, Khwaja and Mian (2005) confirm that despite higher default rates, politically-connected firms receive more funds.

Political instability and civil unrest may curtail financial development. Political instability and corruption may have a detrimental effect on financial development among the sample of low-income countries (Detragiache et al., 2005). Political stability has positive impact on financial development (Aggarwal and Goodell, 2009 and Roe and Siegel, 2013).

Fiscal policy is also of paramount importance for the development of the financial sector. Aside from the potential for inflation, heavily indebted governments may engage in financial repression to use the financial sector as an 'easy' source of funding (Bencivenga and Smith, 1992; Roubini and Sala-i-Martin, 1992; Roubini and Sala-i-Martin, 1995). Excessive public debt would crowd-out private investment, especially in emerging economies with less developed financial sectors (Caballero and Krishnamurthy, 2004; Christensen, 2005).

Maintaining lower inflation is one of the most important national macroeconomic policies which have been documented to be beneficial to financial development. Economies with higher inflation rates are expected to have smaller, less active and less efficient equity markets and banks. Furthermore, inflation raises inflationary expectations and promotes capital outflow and discourages decisions for private activity. Therefore, demand for credit falls. Also, the supply of credit may be negatively affected as a result of a shrinking pool of financial savings since agents

diversify away from liquid assets to keep away from the risk of the inflationary tax. Therefore, it is theoretically expected that inflation hinders financial development. Nonetheless, an alternative theory (Khan, 2002) argues that low levels of inflation on the contrary of the expectation, may foster financial development rather than hindering it. Therefore, inflation may as well have a positive impact on financial development.

Financial development can also be determined by a country's openness to financial inflows. A downside of increased foreign capital flows is the increased risk of capital and current account crises. Financial development is a key factor in ensuring that external shocks do not reverberate within the domestic economy by triggering a financial crisis. The efficient allocation of domestic resources, such as achieving a well-developed stock market, could be beneficial before liberalising the financial systems. Financial opening can only be beneficial when appropriate legal systems and institutions are in place. Financial liberalization is expected to contribute to the efficiency with which markets can transform savings into investments and growth. Hence, according to this view, we should expect higher economic growth, investment and saving rates, as well as financial development following financial liberalization. Overall, it can be concluded that financial liberalization is theoretically expected to lead to financial development.

South Asia receives a huge amount of remittances and official aid from other countries. Although remittances represent a significant portion of the capital inflows in many developing countries their impact on financial development has been under-explored. In theory, to the extent that they are stored in deposit accounts, remittances may expand access to finance and provide unbanked recipients information about other banking products. Billmeier and Massa (2009) show that remittances contribute significantly to stock market development, especially in countries without a sizeable natural-resource endowment. Aggarwal et al. (2011) provide further support, providing strong evidence that remittances improve bank deposits and, to a slightly lesser extent, credit to the private sector, after accounting for a variety of sources for endogeneity and reverse-causality.

Trade flows can also have an impact on financial development. An open economy may weaken the incentives and the political power of interest groups to resist financial deepening (Braun and Raddatz, 2008). Alternatively, increasing exporting opportunities may serve to boost the demand for external funding. Theoretically, trade openness is expected to have an impact on financial development because a raise in the volume of trade increases opportunities for financial

deepening and economic growth. Both these elements are bound to mobilize domestic savings and raise inflows, increasing liquid liabilities in favor of development of financial system. Therefore, capital inflows are also expected to have an impact on financial development because more capital inflows are expected to increase liquid liabilities and support further financial development. From another theoretical standpoint, trade openness encourages economic activity and capital inflows. In support of credit growth, the former channel raises the pool of resources in the financial system. Also, significant increase in credit to the private sector emerges as a result of the latter channel. Equally, credit expansion is a result of capital inflow, which increases available resources in the financial system (Taghipour, 2009).

Rajan and Zingales (2003) interest groups theory argues that politics matter for financial development. They highlight the influence of political systems on financial development policies and point out that closed political systems are more likely to impede the development of financial systems that promote competition and threaten entrenched powers than open political systems. They also highlight the supply-side role of interest groups, especially the vested interests of incumbent industrialists and financial intermediaries. In this view, incumbents have strong motivations to resist financial development because they are worried by the threat of new entrants into the market. In fact, if a country becomes more open to foreign competition or to international flows of capital, these incentives are weakened. Indeed, closed political systems are more likely to impede the development of financial systems that promote competition and threaten entrenched powers than open political systems.

Economic growth is considered another important determinant of financial development. Greenwood and Jovanovic (1990) explain that as the economy grows the costs of financial intermediation falls because of increased competition, which results in an increase in funds available for productive investments. Levine (2005) emphasizes that development of financial sector ought to be in place to drive economic growth. This is because; growth leads to promote development of the financial system and provides motivation to deepen and to widen the system for financial intermediation. Particularly, during periods of economic expansion, the financial sector is more developed; showing financing needs for further development as reaction to real activity (Shaw and Gurley, 1967). That is, because of increased demand for financial services with increased per capita income, expansion of the financial system will be encouraged. Robinson's (1952) hypothesis states that when an economy expands, more financial products, financial institutions and services will emerge in response to larger demand for financial

services. The cost of financial services involves a significant fixed component therefore with increasing the volume of transactions, average costs will fall. The most important theory which explains the impact of economic growth on financial development is the demand-driven hypothesis, according to which the growth of an economy will generate new demand for financial services. Such increase in demand in return, will result in further sophisticated financial intermediaries capable to meet the new demand for their services (Yartey, 2008).

Therefore, from a theoretical perspective, potential determinants of financial development can be listed as trade openness, economic growth, financial liberalization, international financial flows, political rights and inflation. Some of these variables are expected to have a positive impact and some are expected to have a negative impact on financial development. For instance, trade openness, economic growth, financial liberalization and international financial flows are theoretically expected to have a positive impact on financial development whereas inflation is expected to have a negative. Therefore, in light of the theories pointed out in this section, it is expected that economic growth, trade openness and capital inflows, financial liberalization have a positive impact on financial development and Inflation has a negative impact on financial development.

4. Specification of Empirical Model

This study examines the determinants of financial development in South Asian emerging countries. Following Chin and Ito (2005), Huang (2006), Seetanah et al. (2010), and Ayadi et al (2013) the empirical model for financial development can be specified as:

$$FD_{i,t} = \alpha + \beta FD_{i,t-s} + \gamma X_{i,t} + \eta_t + \mu_i + \varepsilon_t$$

Where, FD is the index financial development which includes domestic credit provided by financial sector, domestic credit to private sector and broad money as a percentage of GDP, X is the vector of control variables which includes per capita GDP growth, inflation rate, trade-GDP ratio, remittance-GDP ratio, foreign direct investment-GDP ratio, net official development assistance GDP ratio, financial openness index and political rights index and s is the number of lag years. η_t implies time specific effect and μ_i indicates unobserved country-specific effect. ε_t is random error term. List of variables, their definitions and sources are given in Table 1.

Table 1: List of Variables, their definitions and sources

Domestic credit provided by financial sector (% of GDP)	DCFS	Domestic credit provided by the financial sector includes all credit to various sectors on a gross basis, with the exception of credit to the central government, which is net.	World development Indicators
Domestic credit to private sector (% of GDP)	DCPS	Domestic credit to private sector refers to financial resources provided to the private sector by financial corporations, such as through loans, purchases of nonequity securities, and trade credits and other accounts receivable, that establish a claim for repayment.	World development Indicators
Money and quasi money (M2) as % of GDP	M2	Money and quasi money comprise the sum of currency outside banks, demand deposits other than those of the central government, and the time, savings, and foreign currency deposits of resident sectors other than the central government.	World development Indicators
GDP per capita growth (annual %)	GDP	Annual percentage growth rate of GDP per capita based on constant local currency. Aggregates are based on constant 2005 U.S. dollars. GDP per capita is gross domestic product divided by midyear population.	World development Indicators
Inflation, GDP deflator (annual %)	INF	Inflation as measured by the annual growth rate of the GDP implicit deflator shows the rate of price change in the economy as a whole.	World development Indicators
Trade (% of GDP)	Trade	Trade is the sum of exports and imports of goods and services measured as a share of gross domestic product.	World development Indicators
Remittances, received (% of GDP)	REM	Personal remittances comprise personal transfers and compensation of employees.	World development Indicators
Foreign direct investment, net inflows (% of GDP)	FDI	Foreign direct investment are the net inflows of investment to acquire a lasting management interest (10 percent or more of voting stock) in an enterprise operating in an economy other than that of the investor.	World development Indicators
Net ODA received (% of GNI)	ODA	Net official development assistance (ODA) consists of disbursements of loans made on concessional terms (net of repayments of principal) and grants by official agencies of the members of the Development Assistance Committee (DAC), by multilateral institutions, and by non-DAC countries to promote economic development and welfare in countries and territories in the DAC list of ODA recipients.	World development Indicators
Financial openness index	FOI	It is an index measuring a country's degree of capital account openness	Chinn-Ito's database
Political rights index	PRI	Political right index measures the degree of freedom in the electoral process, political pluralism and participation and functioning of the government.	Freedom House Foundation

To assess the strength of the relationship between financial development and growth, we control for other potential determinants of growth in our regressions. We include log of the initial per capita real GDP to control for convergence and secondary school enrolment ratio to control for human capital accumulation (Beck and Levine, 2004). In addition to these variables, we include consumer price inflation, government consumption expenditure, net inflow of foreign direct

investment, financial openness index, exports and imports. Government consumption expenditure, foreign direct investment, exports and imports are expressed as a percentage of GDP.

5. Data and Methodology

5.1. Data

Financial development is a process that marks improvement in quantity, quality, and efficiency of financial intermediary services. We use domestic credit provided by financial sector, domestic credit to private sector and broad money as a percentage of GDP to measure financial development. A number of explanatory variables are used as determinants of financial development. We use per capita GDP growth, inflation rate, trade-GDP ratio, remittance-GDP ratio, foreign direct investment-GDP ratio, net official development assistance GDP ratio, financial openness index and political rights index as explanatory variables. Table 1 provides summary statistics of dependent variables, i.e., measures of financial development and regressors i. e., determinants of financial development for the study. Most of the data are collected from World Development Indicators 2014, financial openness index data are collected from Chinn-Ito's database and political rights index data are taken from Freedom House Foundation. Table 2 provides the summary statistics of the data that are used in the study.

Table 2: Summary Statistics of Variables over the period 1976-2013

Year	Bangladesh		India		Pakistan		Sri Lanka	
	Mean	Sta dev	Mean	Sta dev	Mean	Sta dev	Mean	Sta dev
DCFS	32.67	16.08	51.01	12.17	48.25	5.16	40.69	6.64
DCPS	21.11	11.89	29.62	10.71	24.23	3.06	23.54	7.28
M2	32.99	16.39	50.36	15.60	42.57	3.27	34.04	4.98
GDP	2.71	1.89	3.85	3.00	2.22	1.91	3.97	1.96
INF	6.90	6.83	7.55	2.87	9.71	4.90	10.90	4.64
Trade	28.60	9.70	25.75	14.39	33.80	2.83	70.40	9.62
REM	4.58	3.07	1.99	1.07	4.99	2.26	5.99	2.24
FDI	0.33	0.42	0.69	0.86	0.86	0.85	1.05	0.59
ODA	4.17	2.30	0.53	0.36	2.32	1.26	5.23	3.36
FOI	-1.41	0.46	-1.18	0.00	-1.19	0.11	-0.32	0.78

6.2. Econometric Methodology

6.2.1. The Panel Data Model

This study uses panel data model to estimate determinants of financial development in South Asian emerging countries. Use of panel data model in estimating common relationships across countries is particularly appropriate because it allows the identification of country-specific effects that control for missing or unobserved variables. The standard form of a panel data model with $i = 1, \dots, N$ and $t = 1, \dots, T$ is given as:

$$Y_{it} = \beta_0 + X'_{it}\beta + \varepsilon_{it} \quad (1)$$

where, X_{it} is a K - dimensional vector of explanatory variables without a constant term, β_0 is the intercept coefficient, β is a $(k \times 1)$ vector of slope coefficient which is independent of i and t , and ε_{it} is the error term that varies over i and t .

6.2.2. Pooled OLS Regression Model

We start with the least squares regression. If we disregard the space and time dimension of our data series, that is, intercept and slope coefficients are constant, we obtain a pooled regression or simply known as OLS regression. Assuming error terms are identically and independently distributed and they are not correlated with exogenous variables, that is, $E(\varepsilon_{it}) = 0$ and $E(X_{it}\varepsilon_{it}) = 0$, we may get consistent estimates. Since individual I is repeatedly observed in the model, the problem of autocorrelation in the error terms arises, i.e., $Cov(\varepsilon_{i,s}, \varepsilon_{i,t}) \neq 0$ with $s \neq t$. This gives inefficient outcome of OLS estimation procedure.

6.2.3. Fixed Effects Model

If we account the individuality of each cross-sectional unit and let intercept vary for each cross-sectional unit but still assume that the slope coefficients are constant across cross-section, the model which we obtain is known as fixed effect regression model. The fixed effects model can be written as:

$$Y_{it} = \beta_{0i} + X'_{it}\beta + \varepsilon_{it} \quad (2)$$

We can allow for the fixed effect intercept to vary between cross-sectional units by following dummy variable technique, thus, incorporating differential intercept dummies in the model. The model can be expressed as:

$$Y_{it} = \beta_{0i}D_i + X'_{it}\beta + \varepsilon_{it} \quad (3)$$

where D is the dummy variable for different cross-sectional units. Through incorporating dummy variables the unobserved heterogeneity problem disappears. The fixed effects model concentrates on the differences within cross sections. It explains to what extent Y_{it} differs from $\bar{Y}_i$. It does not explain why $\bar{Y}_i$ is different from $\bar{Y}_j$ where i and j are two different cross-sectional units. The estimator of fixed effects model is referred to as the covariance estimator or the within-cross-sectional estimator because only the variation within each group is used in forming the estimator.

6.2.4. Random Effects Model

In random effects model individual specific effects and time specific effects are treated random. This model is sometimes referred to as a variance or error components model (ECM). In ECM, it is assumed that the intercept of an individual unit is a random drawing from a much larger population with a constant mean value. The individual intercept is then expressed as a deviation from this constant mean value. One advantage of ECM over FEM is that it is economical in degrees of freedom, as we do not have to estimate all cross-sectional intercepts. ECM is appropriate in situation where the random intercept of each cross-sectional unit is uncorrelated with the regressors. The basic equation of a random effects model can be written as:

$$Y_{it} = \beta_{0i} + X'_{it}\beta + \varepsilon_{it} \quad (4)$$

Now instead of treating β_{0i} as fixed we assume that it is a random variable with a mean value of β_0 (no subscript i here). And the intercept value of a single cross-sectional unit can be expressed as:

$$\beta_{0i} = \beta_0 + u_i \quad i = 1, 2, \dots, N \quad (5)$$

Where u_i is a random error with a mean value of zero and variance of σ_u^2 . Substituting equation (5) into (4) we obtain:

$$Y_{it} = \beta_0 + X'_{it}\beta + \varepsilon_{it} + w_{it} \quad (6)$$

Where w_{it} is a composite error term, i.e., $w_{it} = u_i + \varepsilon_{it}$, u_i is the cross-section error component and ε_{it} is the combined time series and cross-section error component. The term error components model derives its name because the error term w_{it} consists of two (or more) error components. The ECM assumes that the individual error components are not correlated with each other and are not autocorrelated across both cross-section and time series units. The estimation of this type of model requires implementing a more complicated Generalized Least Squares (GLS) procedure than the simpler OLS procedure.

6.2.5. Dynamic Panel Data Model

A panel model is known as dynamic panel model when it uses the lags of the dependent variable as explanatory variables to measure the past realizations of dependent variable. Although the coefficients on lagged dependent variables might be far from our interest, the introduction of these lags becomes crucial to control for the dynamics of the process. Very often allowing dynamic processes is very crucial for recovering consistent estimates of other parameters. Panel data is now widely used to estimate dynamic econometric models. Its advantage over cross-section data in this context is obvious: we cannot estimate dynamic models from observations at a single point in time, and it is rare for single cross-section data set to provide sufficient information about earlier time periods for a dynamic relationship to be investigated. Its advantages over aggregate time series data include the possibility that underlying macroeconomic dynamics may be obscured by aggregation biases and the scope that panel data offers to investigate heterogeneity in adjustment dynamics between different types of cross-sectional units.

The basic form of a dynamic panel data model that contains explanatory variables X_t as well as the lagged endogenous variable Y_{t-1} can be written as:

$$Y_{it} = \rho Y_{i,t-1} + \beta_{0i} + X'_{it}\beta + \varepsilon_{it} \quad (7)$$

where $\varepsilon_{it} \sim N(0, \sigma_\varepsilon^2)$ and $|\rho| < 1$ with $i = 1 \dots N$ index for individuals and $t = 1 \dots T$ index for time periods; X'_{it} is the row vector of explanatory variables with k dimension; ρ is the unknown parameter of the lagged endogenous variable; β is the unknown parameter vector of the k explanatory variables; β_{0i} is the individual specific fixed effects. We assume that the error term is orthogonal to the exogenous variables, i.e., $E(X'_{it}\varepsilon_{it}) = 0$; and that the exogenous variables might be correlated with the individual effects, i.e., $E(X'_{it}\beta_{0i}) \neq 0$. The error term is uncorrelated with the lagged endogenous variable: $E(Y_{i,t-1}, \varepsilon_{it}) = 0$

6.2.6. The SGMM Estimation Technique

Blundell and Bond (1998) show that in case of persistent explanatory variables, delayed levels are weak instruments for difference equation regression proposed by Arellano-Bond estimator. Asymptotically, there may have an increase of the variance of coefficients. In short samples, Monte Carlo simulation shows that weaknesses of instruments can produce biased coefficients. To reduce the potential of the way and the indistinctness associated with the GMM difference estimator, Arellano and Bover (1995) and Blundell and Bond (1998) suggest using a GMM system estimator which combines difference regression with level regression. So the combination of moment restrictions for the differences and levels results in an estimator which is called GMM-system estimator by Blundell-Bond estimator.

The Blundell-Bond estimator is similar to the estimator suggested by Anderson and Hsiao and Arellano-Bond but exploits additional moment restrictions for the differences and levels, which enlarges the set of instruments. These are instruments suited under the additional suppositions. Although it can have a correlation between the levels of the variables of the right side and the specific effect country in the equation (7), there is no correlation between the differences of these variables and the specific effect. Given that the delayed levels are used as instruments in difference regression, the most recent difference is used as an instrument in level regression. The use of delays of additional differences would succeed in the conditions of moment superfluous, (Arellano and Bover, 1995). The dynamic equation to be estimated in levels is given in equ. (7):

$$Y_{it} = \rho Y_{i,t-1} + \beta_{0i} + X'_{it}\beta + \varepsilon_{it} \quad (7)$$

where differencing eliminates the individual effects of β_{0i} and we get the following equation:

$$Y_{it} - Y_{i,t-1} = \rho(Y_{i,t-1} - Y_{i,t-2}) + (X'_{it} - X'_{i,t-1})\beta + \varepsilon_{it} - \varepsilon_{i,t-1} \quad (8)$$

For each year we now look for the instruments available for instrumenting the difference equation. For $t = 3$ the equation to be estimated is:

$$Y_{i3} - Y_{i,2} = \rho(Y_{i2} - Y_{i1}) + (X_{i3} - X_{i2})\beta + \varepsilon_{i3} - \varepsilon_{i2} \quad (9)$$

Where it is assumed the instruments are available. Similarly we can enlarge the instrumentation and for the equation in the final period T we get:

$$Y_{iT} - Y_{i,T-1} = \rho(Y_{i,T-1} - Y_{i,T-2}) + (X_{iT} - X_{i,T-1})\beta + \varepsilon_{iT} - \varepsilon_{i,T-1} \quad (10)$$

Assuming the instruments $Y_{i,1}, Y_{i,2}, \dots, Y_{i,T-2}$, and $X_{i1}, X_{i2}, \dots, X_{i,T-1}$ are available. Finally we get the following instrumented equation that is given in matrix vector form:

$$\dot{W}F_y = \dot{W}F X_i + \dot{W}F_\varepsilon \quad (11)$$

Although the differencing procedure eliminates the specific country effect, it introduces a new way by construction of new error term, which is correlated to dependent variable delayed. According to the suppositions that the error term (ε) is not serially correlated, and the explanatory variables (X) are weakly exogenous, Blundell-Bond propose the following moment's condition:

$$E[(Y_{i,t-s} - Y_{i,t-s-1})(\beta_{0i} + \varepsilon_{it})] = 0 \text{ for } s = 1 \quad (12)$$

$$E[(X_{i,t-s} - X_{i,t-s-1})(\beta_{0i} + \varepsilon_{it})] = 0 \text{ for } s = 1 \quad (13)$$

Hence, we use the moment conditions presented in equations (12) to (13) and employ the system GMM panel estimator to generate consistent and efficient parameter estimates.

One limitation of using the OLS estimation technique in a dynamic panel model is that the method does not account for country specific effects. In order to account for such country specific effects, it is advisable to use the random effect model which can help to capture the qualitative and unobserved effects in a model. There is a problem of unobserved heterogeneity in random effect model which can give biased result. Fixed effect model can handle this issue. The fixed effect model removes all the omitted variables that are constant overtime and hence solve the possible bias that would have resulted if omitted variables are correlated to the included explanatory variables. But fixed effect model suffers from the problem of endogeneity. This problem happens when there is a bi-directional causal relationship between the dependent variable and its regressors. This problem often results in misspecified models which lead to biased estimations. There are at least two sources of endogeneity that may bias estimates of how

the explanatory variables affect the dependent variable in the specified models: first is the unobservable heterogeneity (which arises if there are unobservable factors that affect both the dependent and explanatory variables) and, second is simultaneity (which arises if the independent variables are a function of the dependent variable or expected values of the dependent variable). One other likely important source of endogeneity often overlooked in most empirical researches arises because of the fact that the relations among individual's or country's (in this case) observable characteristics are likely to be dynamic. That is, a country's current performance will affect her future realizations, which will in turn affect her future performance. So, while OLS estimation may be biased because it ignores unobservable heterogeneity, fixed-effects estimation may be biased since it ignores dynamic endogeneity.

The problem of endogeneity that is often associated with the use panel data analysis is thus resolved by the choice of the system GMM estimator to assess determinants of financial development. This methodology not only eliminates any bias that may arise from ignoring dynamic endogeneity, but also provides theoretically based and powerful instruments that account for simultaneity while eliminating any unobservable heterogeneity. Presbitero (2006) provides the evidence that the System GMM produces results that lie between the upper and lower bound represented by OLS and FEM, that show an efficiency gain, and that have had valid instrument set. Therefore, we produce empirical results of system GMM along with results obtained from pooled OLS regression and FEM model.

7. Empirical Results

Results of determinants of financial development are reported in Table 2 to Table 4. In each table, we present results of SGMM methods along with pooled OLS and FEM method. We provide determinants of financial development in terms three indices of financial development.

7.1. Determinants of domestic credit provided by financial sector

Table 3 gives the determinants of domestic credit provided by financial sector. The value of F-statistic in pooled OLS method is found to be 361.39 with a probability value of 0.000 which indicates that the regressors in the model jointly and significantly explain the determinants of domestic credit provided by financial sector. Adjusted R^2 value of 0.9463 shows the model fits well. In pooled OLS model, we find that inflation rate negatively and significantly determine

domestic credit provided by financial sector which implies that financial development requires low level of inflation in South Asian emerging countries. Financial development is positively and significantly associated with its initial indicators in all estimation results. Among the international financial flows, remittance inflows positively and significantly determine the financial development. Though trade-GDP ratio and foreign direct investment have positive impact, their coefficients are found to be insignificant. Theoretically, contradictory signs are found to be for per capita GDP growth rate and net official development assistance GDP ratio, but their coefficients are statistically insignificant.

Table 3: Determinants of Domestic Credit provided by Financial Sector

	Pooled OLS Results		FEM ¹ Results		SGMM Results	
Variable Name	Coefficients	P. Value	Coefficient	P. Value	Coefficient	P. Value
Lag Domestic Credit provided by Financial Sector	0.9029	0.0000	0.7937	0.0000	0.8087	0.0000
GDP per capita growth (annual %)	-0.0228	0.8400	0.0097	0.9310	0.0286	0.7770
Inflation Rate	-0.2016	0.0000	-0.1534	0.0060	-0.1810	0.0000
Trade-GDP Ratio	0.0194	0.4310	0.1313	0.0000	0.0632	0.0120
Remittance-GDP Ratio	0.1927	0.0050	0.3756	0.0000	0.3997	0.0000
Foreign Direct Investment GDP Ratio	2.9533	0.2140	0.3780	0.4130	0.6394	0.1240
Net Official Development Assistance GDP Ratio	-0.1503	0.2340	-0.0532	0.7230	-0.1930	0.1370
Financial Openness Index	-0.3503	0.6040	-0.1730	0.8080	-0.7484	0.2120
Political Rights Index	-0.5124	0.0100	-0.4176	0.8080	-0.6744	0.0010
Constant	6.8006	0.0000	4.9336	0.0080	7.8384	0.0000
Number of Observations	185		185		185	
Number of Cross sections	5		5		5	
Adjusted R ²	0.9463					
F-Statistic/Wald Statistics	361.39	0.0000	290.22	0.0000	3244.74	0.0000

Results imply that FEM method is fit well as is indicated by the F statistics that indicate that explanatory variables in the model jointly and significantly determine dependent variable, domestic credit provided by financial sector. Results reveal that initial value of domestic credit provided by financial sector, trade-GDP ratio, remittance-GDP ratio positively and significantly

¹ In choosing between Random Effects Model (REM) and FEM, we run Houseman specification test and found that FEM is appropriate method.

impact on financial development, whereas, inflation rate and political rights index negatively and significantly impact on financial development. In FEM, the coefficient of per capita GDP growth is found to be positive but insignificant.

Results of SGMM method indicate that domestic credit provided by financial sector is jointly and significantly determined by the regressors. Results report that initial values of domestic credit provided by financial sector, trade-GDP ratio, remittance-GDP ratio have a significant positive impact; whereas inflation rate, political rights index have significant negative impact on domestic credit provided by financial sector. Per capita GDP growth rate and foreign direct investment GDP ratio have positive but insignificant impact on financial development. Net official development assistance GDP ratio and financial openness index have had negative and insignificant impact on financial development.

7.2. Determinants of Domestic Credit to Private Sector

Determinants of second indices of financial development, namely domestic credit to private sector are provided in Table 4. This Table also report estimation results applying SGMM method along with pooled OLS method and FEM method.

Results of pooled OLS method are reported in column 2 and column 3 of Table 4. Estimated value of adjusted R^2 (0.944) shows the model fits well and F-statistic (348.03) shows that its overall fitness is also well. We find that initial value of domestic credit to private sector and remittance-GDP ratio have positive and significant impact whereas inflation rate and political rights index have negative and significant impact on domestic credit to private sector. Per capita GDP growth rate and official development assistance to GDP ratio show positive but insignificant impact whereas, foreign direct investment to GDP ratio and trade GDP ratio show negative but insignificant impact on domestic credit to private sector. One interesting result is found for financial openness index which shows a positive but insignificant impact on domestic credit to private sector implying that strong legal and political institutions, good democratic institutions have positive impact on domestic credit to private sector.

Again, on the basis of Houseman specification test, we consider FEM method as an appropriate one to find out determinants of domestic credit to private sector. Estimated results of FEM method are given in fourth and fifth column of Table 3. Significant value if F-statistic in FEM

method shows that its overall fitness is fairly good. In FEM method, we see that initial value of domestic credit to private sector, trade-GDP ratio and remittance GDP ratio have significant and positive impact on domestic credit to private sector, whereas inflation rate has negative and significant impact on domestic credit to private sector. Per capita GDP growth rates, net official development assistance to GDP ratio and financial openness index have positive but insignificant impact on domestic credit to private sector. Foreign direct investments to GDP ratio and political rights index have negative but insignificant impact on domestic credit to private sector.

Table 4: Determinants of Domestic Credit to Private Sector

Variable Name	Pooled OLS Results		FEM Results		SGMM Results	
	Coefficient	P. Value	Coefficient	P. Value	Coefficient	P. Value
Lag Domestic Credit to Private Sector	0.9244	0.0000	0.8187	0.0000	0.8700	0.0000
GDP per capita growth (annual %)	0.1310	0.1540	0.1093	0.2220	0.1434	0.0690
Inflation Rate	-0.1604	0.0000	-0.1143	0.0090	-0.1590	0.0000
Trade-GDP Ratio	-0.0006	0.9770	0.1046	0.0000	0.0496	0.0180
Remittance-GDP Ratio	0.1954	0.0050	0.3297	0.0000	0.3199	0.0000
Foreign Direct Investment-GDP Ratio	-0.0365	0.9260	-0.0147	0.9690	0.1864	0.5850
Net Official Development Assistance-GDP Ratio	0.0012	0.9900	0.0158	0.9020	0.0487	0.6300
Financial Openness Index	0.3777	0.5010	0.8193	0.1550	0.3653	0.4340
Political Rights Index	-0.3478	0.0430	-0.1058	0.5880	-0.3168	0.0620
Constant	4.2936	0.0010	1.0958	0.4490	2.4481	0.0360
Number of Observations	185		185		185	
Number of Cross sections	5		5		5	
Adjusted R ²	0.9444					
F-Statistic/Wald Statistics	348.03	0.0000	365.05	0.0000	4796.29	0.0000

Results of SGMM estimation are reported in the sixth and seventh column of Table 4. Estimated value of Wald χ^2 Statistic is found to be 4796.29 with 0.000 probability value. It indicates that regressors in the model jointly and significantly determine the determinants of domestic credit to private sector. The coefficient of per capita GDP growth rate is found to be 0.1433542 with 6.9

percent level of significance implying that economic growth plays a positive role to domestic credit to private sector. This finding supports Levine (2005) who emphasize that development of financial sector is crucial to drive growth. This is because growth leads to promote development of the financial system and provides motivation to deepen and to widen the system for financial intermediation. The result suggests that, due to increased demand for financial services with increased per capita income, expansion of the financial system has been fostered in the case South Asian emerging countries. This is indeed in line with Robinson's (1952) hypothesis, which states that when an economy expands, more financial institutions, financial products and services will emerge in response to greater demand for financial services.

We find inflation rate and political rights index have significantly negative impact on domestic credit to private sector, whereas, initial values of domestic credit to private sector and remittance-GDP ratio have significantly positive impact on domestic credit to private sector. Trade GDP ratio significantly and positively impact on domestic credit to private sector. This suggests that an increase in the volume of trade in the case of Iran increases opportunities for financial deepening and economic growth. From another theoretical standpoint, trade openness encourages economic activity and capital inflows, which expands the pool of resources in the financial system. Financial openness index, net official development assistance GDP ratio and foreign direct investment GDP ratio have positive but insignificant impact on domestic credit to private sector.

7.3. Determinants of Broad Money as a Percentage of GDP

Results of pooled OLS, FEM and SGMM methods to evaluate the determinants of third index of financial development, broad money as a percentage of GDP, are presented in Table 5. Results from all three methods imply that all regressors jointly and significantly affect broad money as a percentage of GDP as indicated by significant F statistics. Pooled OLS results show that lagged values of broad money as a percentage of GDP and remittance-GDP ratio have positive and significant impact on broad money as a percentage of GDP. Per capita GDP growth rate has negative and significant coefficient which indicates that per capita GDP growth rate has a negative impact on broad money as a percentage of GDP. Inflation rate and political rights index have negative and significant impact on broad money as a percentage of GDP. Though, trade-GDP ratio and net official development assistance- GDP ratio have positive impact, and foreign direct investment-GDP ratio and financial openness index have had negative impact, although

they are reported to be statistically insignificant. We select FEM model between REM and FEM on the basis of Houseman specification test.

Table 5: Determinants of Broad Money as a Percentage of GDP

Variable Name	Pooled OLS Results		FEM Results		SGMM Results	
	Coefficient	P. Value	Coefficient	P. Value	Coefficient	P. Value
Lag Broad Money as a Percentage of GDP	0.9961	0.0000	0.8756	0.0000	0.9666	0.0000
GDP per capita growth (annual %)	-0.1462	0.0580	-0.1315	0.0850	-0.2128	0.0010
Inflation Rate	-0.2677	0.0000	-0.2392	0.0000	-0.2663	0.0000
Trade-GDP Ratio	0.0022	0.8890	0.1066	0.0000	0.0219	0.2010
Remittance-GDP Ratio	0.1186	0.0180	0.3188	0.0000	0.1841	0.0000
Foreign Direct Investment GDP Ratio	-0.0098	0.9760	0.2868	0.3800	0.3477	0.2020
Net Official Development Assistance GDP Ratio	0.0965	0.2000	0.1126	0.2740	0.0980	0.2330
Financial Openness Index	-0.1348	0.7610	-0.0325	0.9440	-0.3149	0.3980
Political Rights Index	-0.3556	0.0100	-0.1097	0.4970	-0.3260	0.0160
Constant	4.0436	0.0000	2.3469	0.0560	3.7696	0.0000
Number of Observations	185.0000		185.0000		185.0000	
Number of Cross sections	5.0000		5.0000		5.0000	
Adjusted R ²	0.9781					
F-Statistic/Wald Statistics χ^2	912.15	0.0000	812.29	0.0000	11680.00	0.0000

In FEM results, we find the value of F-statistic of 812.29 with probability value of 0.000 indicating that broad money as a percentage of GDP is jointly and significantly determined by regressors. Coefficients of initial values of broad money as a percentage of GDP, trade GDP ratio, remittance GDP ratio are found to be positive and significant implying that these variables have positive impact on broad money as a percentage of GDP. Inflation rate and per capita GDP growth rate have negative and significant impact on broad money as a percentage of GDP. The coefficients of foreign direct investment to GDP ratio and net official development assistance are found to be positive and those of financial openness index and political rights index are found to be negative but they are statistically insignificant indicating that they do not have significant impact on broad money as a percentage of GDP.

Results of SGMM method are given in sixth and seventh column of Table 5. Wald χ^2 statistic is found to be highly statistically significant indicating that all independent variables used in the

model jointly and significantly determine broad money as a percentage of GDP. We find that lagged values of broad money as a percentage of GDP, remittance-GDP ratio significantly and positively impact broad money as a percentage of GDP. As expected, inflation rate has had a negative and significant impact on broad money as a percentage of GDP. Per capita GDP growth rate and political rights index have negative and significant impact on broad money as a percentage of GDP. Trade-GDP ratio, foreign direct investment-GDP ratio and net official development assistance GDP ratio have impacted positively but insignificantly the broad money as a percentage of GDP. Financial openness index does not have impact on broad money as a percentage of GDP in SGMM results.

8. Conclusion

This paper evaluates the determinants of financial development in South Asian emerging countries – Bangladesh, India, Pakistan and Sri Lanka - for the period of 1976 to 2013 applying SGMM as well as pooled OLS and FEM technique. Results obtained from all methods imply that inflation rate shows a negative impact on financial development. This implies that maintaining lower inflation should be important macroeconomic policies in South Asian emerging countries. Results exhibit that per capita GDP growth rate shows a positive impact on domestic credit to private sector, negative impact on broad money as a percentage of GDP and no impact on domestic credit provided by financial sector. Positive growth impact on domestic credit to private sector supports demand driven hypothesis and this indicates that growth of these economies generate new demand for financial services. Such an increase in demand, in return, results in further sophisticated financial intermediaries capable of meeting new demand for their services. Trade GDP ratio is observed to have positive impact on financial development suggesting that a rise in the volume of trade in South Asian emerging countries leads to the increase in the level of financial development. That is, trade openness encourages economic activity and capital inflows. This, in turn, expands the existing services in the financial system. Remittances are found to have positive impact on financial development among the external financial flows. South Asian emerging countries – Bangladesh, India, Pakistan and Sri Lanka – therefore, should be keen to adopt policies which lead to maintaining lower inflation rate, encouraging remittance inflows, generating economic growth to have boasting impact on financial development and this in turn could bring accelerating effect on economic growth; moreover, keeping trade liberalized and easing the flow of remittances should also be put in the policy options.

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